Lifelong Learning for an Origami Robot

Supervised Learning of Target Adaptations

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by

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***Abstract*— Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.**

I. Introduction

II. Background

A. Lifelong Learning

B. Kresling Origami

III. Methodology

A. Target Adaptations

The method proposed in this paper aims to provide a way of improving a given controller's performance as well as fortifying it against evolving system dynamics *without* having access to the inner workings of the controller itself. In other words, provided that the most base assumptions of a controller hold true—i.e. it requires a state and a target as input and yields a control action—the method can be completely agnostic of the controller that is used.

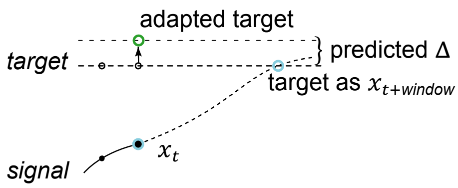

Fig. 4. Target adaptation during deployment: By supplying the model with the controller's target as $x_{t+\text{window}}$ it yields a Δ that can be added to the current target, which will supposedly result in the controller reaching the target within the specified window.

B. Toy Problem

To explore the proposed method, a toy problem is considered first. The toy problem consists of a simulation of a simple 1D mass-spring-damper system and an imperfectly tuned PID controller. With intervals of 5 seconds, a series of random targets are generated for the controller to reach and maintain. As the simulation progresses, the parameters of the dynamic system are slowly changed to simulate a degradation in the system.

C. Preliminary Results

IV. A Kresling Origami Robot

V. Experiments and Results

VI. Improved Model

VII. Discussion